

Hybrid AI with Explainable AI for Soil Microbiome Analysis and Predictive Modeling for Mango Crop at Malihabad (U.P.)

1. Origin of the Proposal

The productivity and sustainability of mango, a high-value horticultural crop, are critically influenced by soil health. Soil microbiomes—complex communities of Archaea, fungus, bacteria, and other microbes—play pivotal roles in pathogen suppression, nutrition cycling, and stress resilience. However, the interactions between microbiomes, soil parameters, and crop yield are highly complex and non-linear, making conventional analysis insufficient for actionable predictions.

Recent advances in metagenomics and bioinformatics have generated large-scale soil microbiome datasets. Yet, the challenge lies in extracting interpretable patterns that connect microbial communities to crop performance. Artificial Intelligence (AI), particularly hybrid approaches that combine Deep learning (DL) and machine learning (ML) provide a revolutionary solution for modeling these complex datasets. The integration of **Explainable AI (XAI)** [1] ensures that predictions are not only accurate but also interpretable, highlighting microbial and soil features most critical for mango productivity.

The identification of three overlapping needs served as the impetus for this proposal:

1. **Scientific:** Bridging microbiome data with crop-specific outcomes through advanced AI models.
2. **Agricultural:** Enhancing mango yield, resilience, and quality under conditions of soil degradation and climate stress.
3. **Societal:** Providing sustainable, data-driven recommendations that reduce reliance on chemical inputs and promote eco-friendly practices.

The specific objectives are to:

- Characterize mango orchard soil microbiomes across diverse agro-climatic zones like Malihabad (U.P.).
- Develop hybrid AI models for predictive analysis.
- Apply explainable AI techniques for interpretability and actionable insights.
- Validate predictions and provide decision-support tools for sustainable mango cultivation.

The novelty of this proposal lies in the integration of **hybrid AI and XAI** [2] with soil microbiome science for a location- Malihabad in U.P. and crop-specific application. The anticipated impact includes higher mango yields, enhanced soil health monitoring, reduced agrochemical use, and contributions to global sustainability goals.

2. State of Art Review

2.1 Review of Status & Significance

Globally, soil microbiome research has gained momentum due to its role in sustainable agriculture and climate resilience. Metagenomic sequencing platforms and bioinformatics pipelines have enabled characterization of soil microbial communities at unprecedented resolution. Recent studies have linked microbial taxa to nitrogen fixation, phosphorus solubilization, organic matter turnover, and disease suppression. In horticultural systems, particularly perennial crops like mango, microbiomes play crucial roles in long-term soil fertility and fruit quality.

International Status:

- Studies in the US, Europe, and China have demonstrated predictive links between soil microbial networks and crop yield using machine learning approaches.
- Research on maize, rice, and wheat has successfully applied ML models for yield prediction and disease risk assessment.[3]
- Hybrid AI approaches (combining ML and DL) are emerging in plant microbiome studies, though largely focused on staple crops rather than horticultural systems.
- Explainable AI is being recognized as essential to bridge computational models with biological and agronomic interpretability.

Status in India:

- India is the world's largest producer of mango, yet yield variability remains high due to soil degradation, inconsistent management practices, and climate impacts.
- Existing studies have examined mango orchard soil properties and disease profiles, but comprehensive microbiome-based predictive frameworks are lacking.
- AI applications in Indian agriculture are growing, but explainability and crop-specific microbiome integration remain underexplored.

Research Gaps:

- Lack of crop-specific microbiome prediction models for mango.
- Limited integration of hybrid AI with explainable outputs for actionable soil management.
- Absence of location-specific soil microbiome datasets for mango agroecosystems.
- Need for field-validated predictive models that link microbiome dynamics with yield and sustainability indicators.

2.2 Importance in Terms of Impact & Innovation

This proposal is innovative in three dimensions:

1. **Scientific Innovation:** First-of-its-kind integration of hybrid AI with explainable frameworks in mango-focused soil microbiome analysis.

2. **Practical Impact:** Development of predictive models that provide actionable recommendations for farmers, enhancing yield and reducing input costs.
3. **Policy & Sustainability:** Contribution to soil health monitoring frameworks, supporting national and international goals on sustainable agriculture, biodiversity, and climate action.

2.3 Proof of Concept

Preliminary pilot analyses of publicly available microbiome datasets (e.g., rhizosphere communities in horticultural systems) indicate that ML models can classify soil samples with >80% accuracy for nutrient availability and disease risk. Hybrid approaches have shown further improvement in robustness. While no mango-specific dataset has yet been modeled, these early results provide confidence in the viability and scalability of the proposed approach.

2.4 Location Specificity

Mango cultivation in India spans diverse agro-climatic zones (Uttar Pradesh, Maharashtra, Andhra Pradesh, Karnataka, West Bengal). Soil types vary significantly— from lateritic and black cotton soils in the Deccan Plateau to alluvial soils in North India. This variability makes mango an ideal case study for developing location-specific predictive models. The project will collect soil samples from major mango belts specially from Malihabad (U.P.), ensuring regional representation and locally relevant outcomes.

3A. Work Plan

3.1 Methodology

The goal of the study technique for the project is to methodically combine data analytics, artificial intelligence (AI)-based modeling, and biological testing. The multidisciplinary approach combines soil microbiology, data science, agricultural engineering, and bioinformatics to generate accurate and intelligible forecast findings for sustainable mango farming.

1. Study Design and Sampling Framework

The study will be conducted across major mango-growing regions specially Malihabad in U.P. representing diverse agro-climatic zones. Representative orchards will be selected to capture variability in soil type, management practices, and productivity levels. For each site, associated metadata—such as soil pH, organic carbon, nutrient content (N, P, K), temperature, moisture, and yield data—will be recorded [6]. This stratified sampling design ensures the inclusion of ecological and management variability essential for model generalization.

2. Soil Microbiome Analysis and Data Acquisition

The sequencing data will be processed through a bioinformatics pipeline involving quality filtering, operational taxonomic unit (OTU) clustering, and taxonomic assignment using databases such as SILVA and Greengenes.

3. Data Integration and Preprocessing

Multimodal datasets—including microbiome data, soil physicochemical parameters, weather variables, and agronomic data—will be integrated into a unified data warehouse. Data normalization, missing value imputation, and dimensionality reduction (via PCA or t-SNE) [5,7,8] will be applied to ensure high-quality input for modeling. Feature engineering will be used to generate synthetic variables such as microbial richness scores, nutrient balance ratios, and soil health indices, enhancing the interpretive capacity of the AI models.

4. Development of Hybrid AI Models

Hybrid AI models will be developed to capture the complex, non-linear interactions among soil, microbes, and crop performance. The models will combine:

- **Machine Learning algorithms:** Random Forest, Gradient Boosting, and Support Vector Machines for feature selection and pattern discovery.
- **Deep Learning architectures:** Graph neural networks (GNNs) and convolutional neural networks (CNNs) are used to simulate relational and hierarchical relationships in microbial networks.
- **Rule-based reasoning systems:** Predictions will be guided and constrained by agronomic principles and expert knowledge, improving biological plausibility. Cross-validation techniques will be used to train and test the models, and metrics including accuracy, F1-score, and root mean square error (RMSE) will be used to assess performance.

5. Explainable AI Implementation

In order to guarantee interpretability, Explainable AI (XAI) [2] methods such feature-importance visualizations, SHAP (Shapley Additive Explanations), and LIME (Local Interpretable Model-agnostic Explanations) [4] will be used. These techniques will pinpoint the most significant soil characteristics, microbial taxa, and climate factors influencing forecasts. Transparent decision-making will be supported by attention-based models, which will also help see dependencies and interconnections.

6. Decision Support System (DSS) Development

The development of a web-based decision support system prototype will incorporate explainable outputs and prediction models into an interactive dashboard. Users will be able to enter environmental and soil characteristics into the DSS [3] and get recommendations for yield optimization, microbial augmentation, and soil management that are easy to understand.

7. Validation and Field Trials

To verify the AI-generated predictions in practical settings, field tests will be carried out. The effects of the recommendations on microbial balance, soil fertility, and yield will be evaluated. The models and DSS will be improved by the validated results, guaranteeing scalability and practical applicability across mango-producing regions.

This multi-phase approach will allow the research to bridge the gap between advanced AI modeling and experimental microbiology, producing data-driven, interpretable insights for sustainable soil health management and precision mango farming.

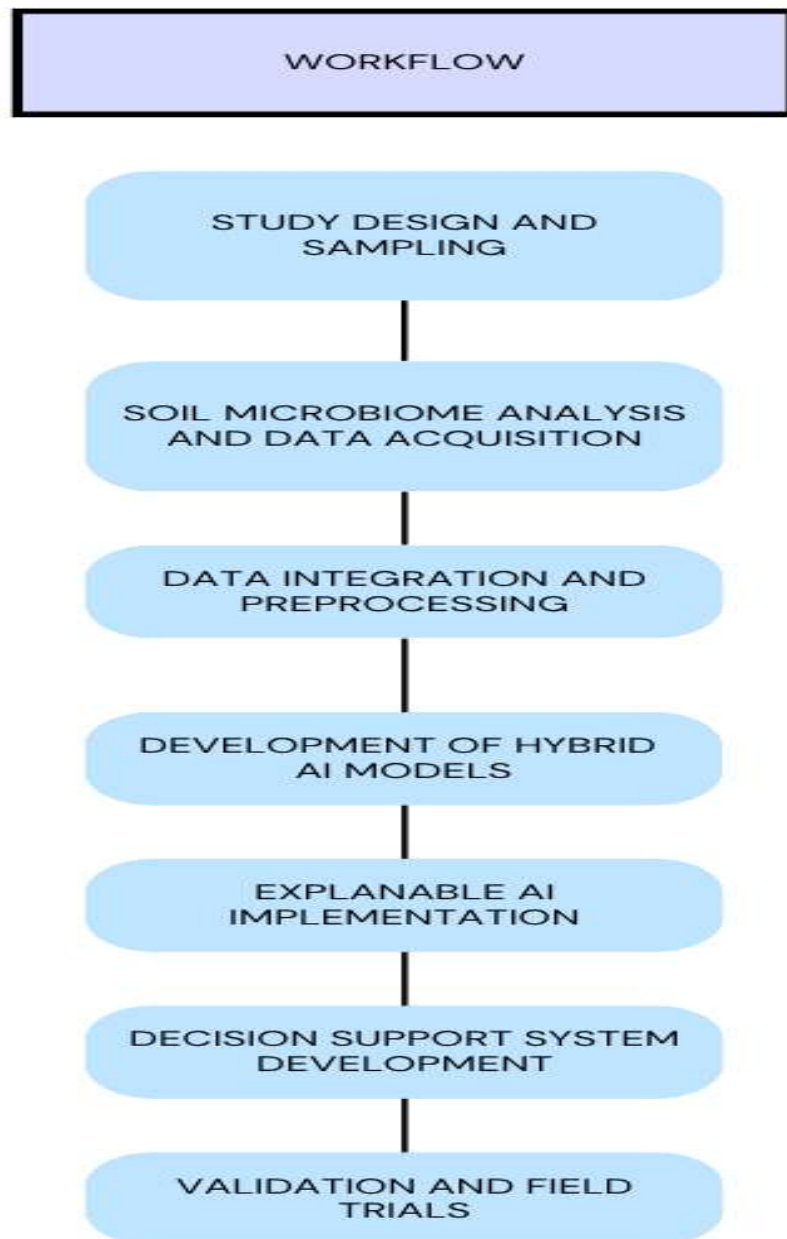


Figure-1

3.2 Time Schedule & Milestones

Task/Phase	M1-3	M4-6	M7-9	M10-12	M13-15	M16-18	M19-21	M22-24
1. Project Planning & Literature Review	✓							
2. Soil & Mango Crop Data Collection		✓	✓	✓				
3. Data Preprocessing & Microbiome Analysis			✓	✓	✓			
4. Hybrid AI Model Development				✓	✓	✓		
5. Explainable AI (XAI) Integration					✓	✓	✓	
6. Predictive Modeling for Mango Yield					✓	✓	✓	✓
7. Validation & Field Testing						✓	✓	✓
8. Documentation & Reports							✓	✓
9. Publications & Knowledge Transfer		↓						✓

Figure-2

3.3 Additional Information

3.3.1 Social/Economic Benefit

- **Farmers:** Improved yields and reduced input costs through data-driven soil management.
- **Economy:** Strengthened mango export potential by ensuring consistent quality and sustainability.
- **Society:** Enhanced food security and farmer livelihoods, particularly in mango-growing regions.

3.3.2 Environmental Impact & Risk Analysis

Public health and environmental safety will improve as microbial and AI-guided interventions reduce soil and water contamination levels and lessen dependency on chemical fertilizers and pesticides. Better dietary quality and food safety will result indirectly from healthier soils producing fruits with higher nutritional value. Microbial diversity will improve biodiversity and environmental conservation initiatives by restoring ecological balance.

3.3.3 Alignment with SDGs and Schemes

This proposal aligns with **SDG 9 (Industry, Innovation, and Infrastructure)**; **SDG 3 (Climate Action)**; **SDG 12 (Responsible Consumption and Production)**; **SDG 15 (Life on Land)**; and **SDG 2 (Zero Hunger)**. It supports India's **National Mission on Sustainable Agriculture**, **National AI Strategy (NITI Aayog)**, and global initiatives such as the **FAO Global Soil Partnership** and the **4 per 1000 Initiative**. The emphasis on explainable AI aligns with **UNESCO's AI ethics framework** and **OECD guidelines**.

4. Expertise

4.1 Expertise of Investigators

Dr. Pragati Upadhyay is the Main Investigator of this project. She has rich experience in Machine Learning applications. She has worked on optimization algorithms in the area of Machine Learning. Her work is in area of Periodic Pattern Mining from Spatio Temporal Database Using Novel Global Pollination Artificial Fish Swarm Optimizer based Clustering and Modified FP Tree.

4.2 Roles & Responsibilities of PI and Co-PI

S. No.	Name of the Investigator	Responsibilities
1	Dr. Pragati Upadhyay	Key Work, Sampling, Preparation of work model
2	Dr. Amit Yadav	Verification and Publication work

4.3 Key Publications (Last 5 Years)

1. U.Trivedi, P.Upadhyay, “**Comparative Analysis of SVM with Logistics Regression and Decision Tree on Non-Linear Data Points**” in the Seventh International Conference on Advances in Electric and Computer Technologies 2025 (ICAECT 2025). Accepted to be published in Scopus Indexed Journal in the month of December 2025.
2. S. Jha, A. Sanghi, P. Upadhyay and G. Agarwal, "Analysing the Identification Approaches of Deep Fake Images and Videos Encapsulated in Fake Contents Available on Social Platforms," 2024 International Conference on Communication, Control, and Intelligent Systems (CCIS), Mathura, India, 2024, pp. 1-6, doi: 10.1109/CCIS63231.2024.10931974.(**Scopus Indexed**)
3. A.Gaur, B. Kumar, P. Upadhyay, "**Detect the Selfishness of Node in Wireless Sensor Network**" 2025 Tuijin Jishu /Journal of Propulsion Technology (**Scopus Indexed**) ISSN:1001-4055, Vol.46 No.1(2025).
4. P. Upadhyay, J. Sangeethapriya., A. Kumar, R. B. Dhumale, R. Singh and V. Tripathi, "**A Critical Analysis on the Security Attacks and Relevant Countermeasures using ML,**" 2022 5th International Conference on Contemporary Computing and Informatics (IC3I), Uttar Pradesh, India, 2022, pp. 96-101, doi: 10.1109/IC3I56241.2022.10072972 (**IEEE Explore**).
5. P. Upadhyay, M. K. Pandey and N. Kohli, “Periodic Pattern Mining from Spatio Temporal Database Using Novel Global Pollination Artificial Fish Swarm Optimizer based Clustering and Modified FP Tree”, Soft Computing (**SCIE-indexed**), vol. 25, pp. 4327– 4344, 2021.
6. P. Upadhyay, M. K. Pandey and N. Kohli, “Mining Periodic Patterns from Spatio Temporal Trajectories Using FGO-based Artificial Neural Network Optimization Model”, Neural Computing and Applications (**SCIE-indexed**), 2021.

5. Facilities at Host Institute

- CoreTM i-9-10900@2.80GHz, 32 GB Ram, Windows 10 Pro (64 Bit)
- 1TB HDD, 500 GB SSD, GPU NVIDIA GeForce RTX 3070 8GB

6. Experts/Institutions Interested in Project Outcome

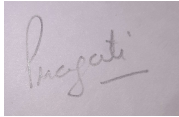
S.No.	Name of Expert Affiliation	Email	Area of Expertise
1	Prof. Brijesh Charausia Professor & Dean Research Computer Science and Engineering Office No B-31, Block B, Dept. of CSE Pranveer Singh Institute of Technology, Kanpur, 209305 INDIA.	brijeshchaurasia@ieee.org	Machine Learning
2	Dr. Man Mohan Shukla Professor Group Director Office No A-14, Block A Pranveer Singh Institute of Technology, Kanpur, 209305 INDIA.	mshukla@psit.ac.in	Machine Learning

7. Budget Estimate

Budget Head	Item Description	Quantity	Unit Cost	Total Cost
A. Equipment	AWS cost			25000
B. Consumables & Chemicals	Sesors	10	2000	20000
C. Travel	6 visits			10000
Manpower	1 research fellow for two years		5000 per month	120000
Contingency	Stationary, printing, publishing			25000

8. Bibliography

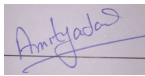
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Signature of PI

Dr. Pragati Upadhyay

Date: 10/10/2025



Signature of Co-PIs

Dr. Amit Yadav

Date: 10/10/2025